

Selective Ablation of Cu-Coated PEEK using Femtosecond Lasers

Mingchen Du¹, Jingbo Liu², Jiancheng Shi², Chao Deng², Mengmeng Wang^{*1}

¹School of Mechanical Engineering, Beijing Institute of Technology, China

² Southwest Electronic Equipment Research Institute of China, China

ABSTRACT

Femtosecond laser direct writing represents a powerful tool for the precise ablation and patterning of Cu/Ni/Au-coated PEEK substrates, essential for lightweight circuits and conformal antennas in aerospace applications. This study investigates the dynamic interaction between femtosecond laser pulses and multilayer materials, examining how variable pulse widths impact the ablation thresholds of copper and PEEK. Results indicate that, while both materials exhibit increased thresholds with longer pulse durations, copper consistently requires a higher threshold, making selective copper ablation challenging at a wavelength of 1030 nm. This inherent difference necessitates careful control to minimize substrate damage, as copper removal typically leads to partial ablation of the underlying PEEK. We implemented an in-situ monitoring system based on Laser-Induced Breakdown Spectroscopy (LIBS), providing real-time compositional and depth feedback during repeated scans. This system enabled precise identification of the copper-PEEK interface, allowing efficient material removal with minimized impact on the substrate. Post-ablation characterization, utilizing SEM, Raman spectroscopy, and XPS, revealed that optimized femtosecond laser parameters effectively preserve the chemical integrity and morphology of PEEK, reducing thermal degradation and carbonization. This work establishes femtosecond laser direct writing as a viable alternative to traditional multi-step processes, offering refined control over ablation depth and precision in polymer-metal composites, with promising implications for advanced aerospace applications.

Keywords: Femtosecond laser, Selective Ablation, Multilayer Materials Processing, Micro-nano processing

1. INTRODUCTION

With the increasing demand for electromagnetic wave transmission control in advanced aircraft, conformal antennas with frequency-selective or polarization-sensitive functions play a vital role in aerospace equipment. These integrated structure/function components unify electromagnetic functionality and structural integrity through precisely designed thin-film patterns, significantly enhancing the aircraft's stealth and communication capabilities. These antenna components are typically composed of dielectric materials coated with Cu, Ni, or Au metal films, among which the specialized polymer PEEK (polyether ether ketone) is an ideal substrate for conformal antennas due to its high strength, corrosion resistance, design versatility, and excellent thermal stability. To ensure strong adhesion of the metal film, the PEEK substrate is generally roughened to improve bonding strength.

During processing, it is essential to selectively remove the metal layer without damaging or carbonizing the underlying PEEK. Current methods include pre-drawing circuit patterns via photoresist exposure and development, followed by electroless copper plating, etching, and stripping processes to remove excess copper foil and photosensitive material. However, this process is complex, environmentally taxing, and limited in resolution. In contrast, femtosecond laser direct-writing patterning technology offers high resolution, flexibility, efficiency, and environmental benefits, making it an ideal approach for optimizing existing processes to meet the high-performance requirements of conformal antennas.

In recent years, researchers worldwide have conducted extensive studies on laser processing of resin-based copper-clad laminates. For instance, Deng et al. examined the large-area copper layer stripping process of FR-4 copper-clad laminates under nanosecond laser induction, utilizing mechanical forces generated by resin decomposition to remove the copper layer.^[1-3] Weber and Graf noted that heat accumulation between laser pulses is a primary factor in the formation of heat-affected zones during processing.^[4] Gaitanelis et al. analyzed the thermal degradation of peek in laser processing.^[5] Researchers have investigated the mechanisms of laser processing in polymers, focusing on energy deposition and its impact on material properties.^[6-11] Some have explored aspects such as laser-induced photochemical and photothermal reactions, energy absorption dynamics, and ablation thresholds, deepening our understanding of how these mechanisms influence surface morphology and structural integrity.^[12-15] However, research on the ultrafast laser processing characteristics of polymer-based copper-clad materials remains limited. Compared to traditional machining and long-pulse or continuous lasers, ultrafast lasers offer significant advantages in damage control and processing accuracy. Given the substantial differences in thermal conductivity and energy absorption mechanisms between the metal laminated film and the PEEK matrix, as well as in processing damage levels and thresholds, it is critical to optimize laser processing parameters to achieve efficient metal layer removal while preserving the PEEK substrate's integrity.

To address the circuit patterning requirements for PEEK-based conformal antennas, this study investigated the ablation threshold for 1030 nm femtosecond lasers using area epitaxy methods and examined the threshold's dependence on pulse width to identify high-quality near-infrared femtosecond laser processing parameters. Additionally, by incorporating laser-induced breakdown spectroscopy (LIBS), in-situ depth monitoring was achieved during processing to ensure the base material was not excessively removed. Finally, post-machining characterization of surface morphology and chemical composition verified the optimal laser parameters necessary to balance machining accuracy and substrate protection.

2.EXPERIMENTAL DETAILS

In this study, a YactoFiber-FL-20 high-power femtosecond laser, produced by Hangzhou Yacto Technology Co., Ltd., was utilized as the experimental laser source. The laser beam was guided and focused onto the material surface using an Aerotech AGV10HP-BE1 two-dimensional galvanometric scanner equipped with a 160 mm focal length flat-field lens. The sample was positioned at the laser's focal plane, where it underwent cyclic scanning and etching by the Gaussian laser source. The parameters of the linearly polarized pulsed Gaussian laser beam employed in this study are presented in Table 1.

Table 1. Laser and processing parameters

Parameter	Value
Average power P/W	0-20
Repetition frequency f / kHz	100
Pulse duration (1/2 light intensity)/fs	138
Wavelength λ /nm	1030
waist radius $w_0/\mu\text{m}$	21.16
Overlap	88%
Scanning speed $v_s / \text{mm}\cdot\text{s}^{-1}$	500
Scanning pass n	1-25

The sample selected for this study was a high-purity PEEK resin substrate, model 5600G, provided by Jiangsu Junhua Special Polymer Materials Co., Ltd. A composite metal film consisting of Cu10/Ni1/Au0.5 (10 μm Cu, 1 μm Ni, 0.5 μm Au) was deposited onto the PEEK substrate via magnetron sputtering by Zibo Dayou Electronics Technology Co., Ltd. This coated substrate was used as the test material for processing experiments. Its parameters and properties are presented in Table 2.

Table 2. Physical and Thermal Properties of the Sample Material

Property	Specification	Result
Density(g/cm^3)	1.3 $\pm$ 0.05	1.30
Thickness(mm)	4	4
glass transition temperature($^{\circ}\text{C}$)	143 $^{\circ}\text{C}$	-
melting point($^{\circ}\text{C}$)	334 $^{\circ}\text{C}$	-
dielectric constant(ϵ)	3.2	-

3.RESULTS AND DISCUSSION

3.1Ablation thresholds of copper and peek

The ablation threshold is a critical parameter in ultrafast laser processing. It influences structural dimensions, such as width and depth, and provides guidance for optimizing other processing parameters to control quality and efficiency. Additionally, it serves as a key indicator for elucidating the laser-material interaction mechanisms based on observed processing phenomena.

For the processing of the copper clad layer on the PEEK substrate, studying the relative magnitude relationship between the ablation thresholds of copper and PEEK is a necessary step to achieve damage-free processing of the substrate.

First, the single-pulse ablation threshold of PEEK material was measured under a pulse width of 138 fs. Laser single pulses with different energy levels (1–100 μJ) were used to perform single-point ablation threshold measurements on the polished PEEK substrate. Fig.1 shows the typical morphologies of ablation craters created by single-point impacts at varying energy levels.

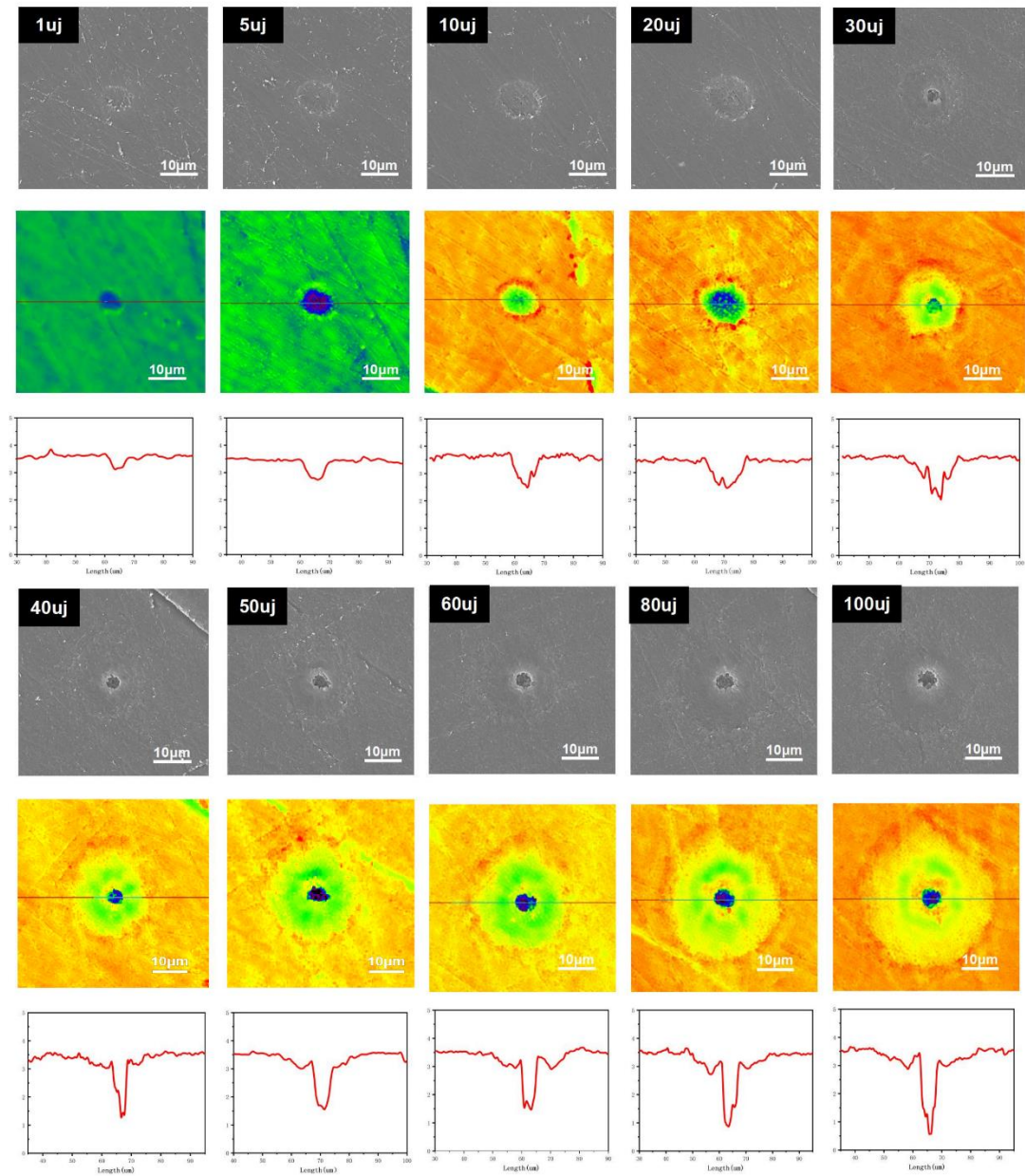


Fig.1. Ablation morphologies at single-pulse energies ranging from 1 to 100 μJ

It can be seen that when the laser fluence is lower than $13.6\text{J}/\text{cm}^2$, the ablation crater is relatively flat and a keyhole appears after femtosecond laser irradiation. At this time, the laser energy fluence just exceeds the ablation threshold of the material. When the laser fluence reaches $20.4\text{J}/\text{cm}^2$, the material exhibits obvious central explosive boiling removal, substrate evaporation zone and crater morphology. With the increase of the fluence, the width of the central ablation and the range of the substrate evaporation zone gradually increase. It can be seen from the morphology that the thermal effect in the femtosecond laser - irradiated area is obvious. Explosive boiling occurs followed by solidification under the condition of far exceeding the threshold, while vaporization zone appears under the near - threshold condition.

The area extrapolation method can be used to determine parameters such as the actual spot radius ω_0 (defined as the "1/e²" spot radius), the ablation threshold F_{th} , and the laser fluence F applied to the material surface. Specifically, according to the theoretical formula of the area extrapolation method, the linear relationship between D^2 and $\ln P$ can be obtained:

$$D^2 = 2\omega_0^2 \ln(P/P_{th})$$

Where P is the average power of the pulse sequence, which can be obtained by multiplying the energy E of each pulse in the sequence by the pulse repetition frequency f ; by performing linear regression on several pairs of $(\ln P, D^2)$ coordinate points using the least squares method, the spot radius ω_0 and the ablation threshold energy E_{th} can be obtained from the slope and intercept of the fitted line, respectively.

After the spot radius ω_0 is obtained, the power value P of each pulse can be converted into the energy fluence F . The relationship between the peak fluence (the fluence at the center of the Gaussian spot) F and the average power P at its position can be used:

$$F = 2P/\pi f \omega_0^2$$

$F_{th} = 0.6806 \text{ J/cm}^2$ was obtained through the extrapolation method.

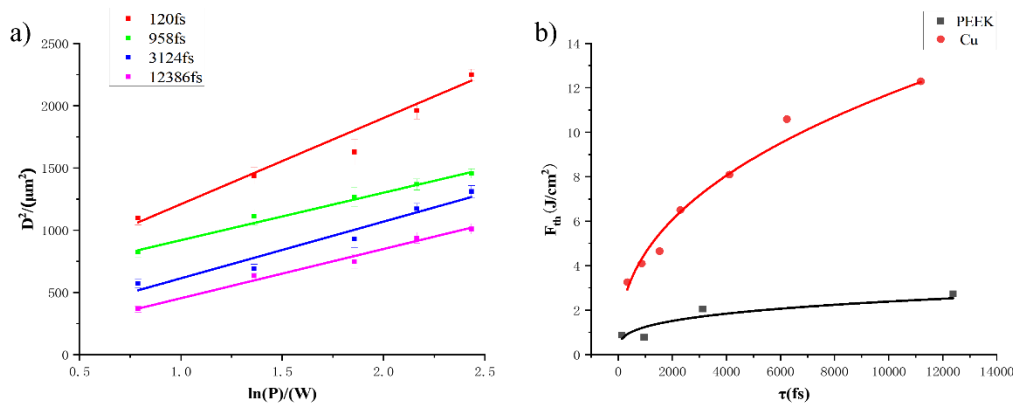


Fig. 2. Ablation Thresholds of PEEK and Copper under Variable Pulse Widths. a) Determination of the ablation threshold for PEEK at varying pulse widths. b) Comparative analysis of ablation thresholds for PEEK and copper across different pulse widths.

For the single-pulse ablation threshold of Cu under short pulse widths, Zayarny research group measured $F_{th} = 1.8 \text{ J/cm}^2$ through experiments^[16]. The necessary condition for achieving damage-free copper clad layer on the PEEK substrate is that the ablation threshold of copper must be lower than that of PEEK. A research group had already studied the ablation threshold of various metals under femtosecond laser ablation at a wavelength of 1030 nm with pulse widths ranging from 0.5 to 20 ps. The pulse width of Cu under infrared laser (1030 nm) increases in the range of 0.5 to 20 ps, and the increase in the threshold is related to two-photon interband (d-s) absorption in Cu.^[17] Similarly, PEEK is a polymer, and its UV-visible absorption spectrum indicates that the optical band gap of PEEK is approximately 3.1 eV.^[18] The absorption at 1030 nm (1.2 eV) depends on three-photon absorption.

Fig.2 demonstrates that while the ablation threshold of PEEK can be controlled by adjusting the pulse width, the ablation threshold of Cu remains consistently higher than that of PEEK. Both materials show an increase in ablation threshold with increasing pulse width. When patterning the copper plating on the PEEK surface, laser ablation at the

copper-PEEK interface will inevitably result in ablation of the substrate. Therefore, whether the ablated PEEK after circuit patterning with 1030 nm fundamental light is prone to carbonization and whether its morphology is controllable are key focuses of our further research.

3.2 Online depth monitoring

In this study, a plasma LIBS observation system was employed to collect spectral and intensity information during plasma expansion in the processing stage. This approach enabled differentiation of the material composition in the scanned area and identification of the material removal depth.

Based on the standard data from the NIST LIBS emission spectrum database, the Cu I excited-state plasma spectrum for copper exhibits emission peaks at 510.57 nm, 515.32 nm, 521.87 nm, 529.87 nm, 570.03 nm, and 578.24 nm. For the PEEK substrate, as a polymer material, laser-induced plasma emissions include monomer radicals, carbides, and minute droplets. Among these emissions, the Swan bands of carbide C_2 can be distinctly identified in Fig.3. The broad spectrum centered at 516 nm corresponds to the C_2 Swan band with $\Delta v = -1$, while the broad spectrum centered at 588 nm corresponds to the $\Delta v = -2$ Swan band, which overlaps with the Na I plasma emission spectrum from sodium impurities within the polymer.^[19]

Using the collected spectral and intensity data, material information within the laser focal region can be identified in the processed area. With the sample's coating thickness fixed at 1 μm , the removal depth remains consistent across laser scans. By calibrating scan counts against coating thickness, effective in-situ depth monitoring is achieved.

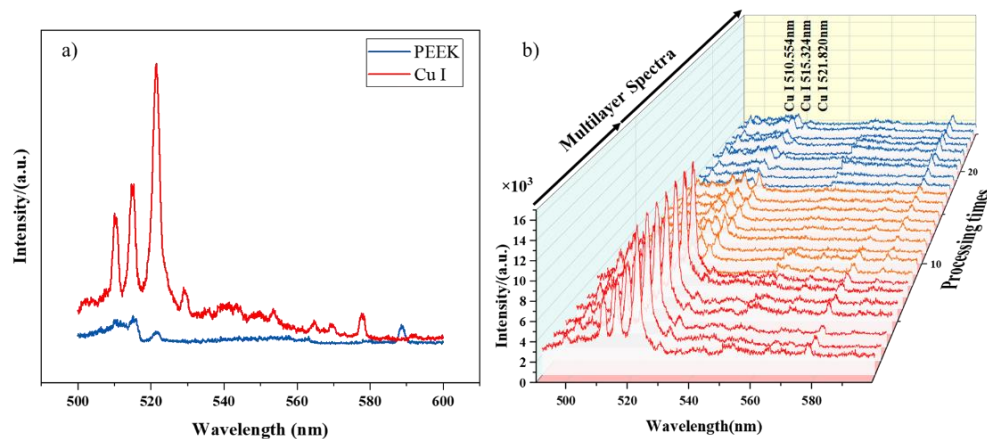


Fig.3. LIBS spectral signals for in-situ depth monitoring (a) LIBS spectra of pure PEEK and copper. (b) In-situ spectral signal evolution during multiple scans.

By distinguishing the 578.24 nm emission line of the Cu I excited-state plasma from the broad Swan band spectrum of the C_2 molecule at 588 nm ($\Delta v = -2$), real-time monitoring of material composition and depth removal is achieved. Spectral data were collected within the 490–600 nm range. As shown in Fig. 3, after 10 scans, the 588 nm line of the C_2 molecule begins to appear, indicating that processing has reached the interface between the copper layer and the PEEK substrate. Due to the substrate's high roughness, regions of incomplete copper removal are still present at this stage,

resulting in simultaneous signals from both copper and PEEK. After 23 scans, the Cu I emission line at 578.24 nm nearly disappears, signifying complete removal of the copper layer.

Although the 515 nm emission line of copper atoms nearly coincides with the 516 nm Swan band ($\Delta\nu = -1$) of C_2 molecules, the copper line exhibits an intensity approximately 10 times greater than that of the substrate signal. This intensity difference allows for additional compositional differentiation using the plasma signal at 515 nm, despite the spectral overlap. These results indicate that both the interface between copper and PEEK and the precise number of scanning layers required for complete copper removal can be accurately identified, thereby minimizing excessive ablation depth into the substrate.

The copper layer was deposited on the sample surface using magnetron sputtering, with an average thickness of 1 μm . Scanning electron microscopy (SEM) characterization aligns closely with the in-situ LIBS spectral data, indicating that the substrate material is exposed after 10 scans, with complete removal of the copper layer after 20 scans. The morphology of the material removal from scan 1 to scan 20 is shown in Figure.4, where the removal of the copper layer reveals the mixed interface of copper and PEEK, leading to a flat PEEK surface.

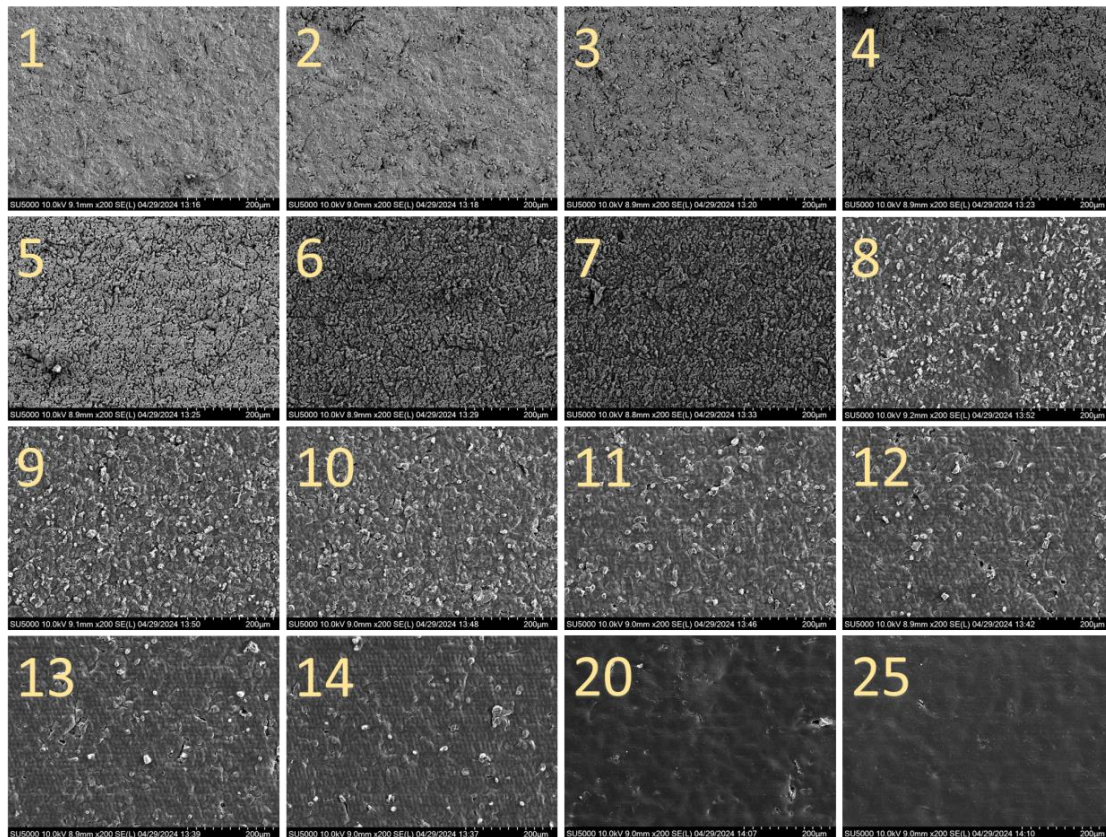


Fig.4. Morphology of the sample under SEM after multiple scans.

Depth information measured with a laser confocal microscope shows that the average removal depth after 5 scans is 1.11 μm , while after 14 scans, it reaches 13.49 μm . The average removal efficiency for the copper layer is 0.222 μm per scan, and for the PEEK substrate, it is 1.37 μm per scan.

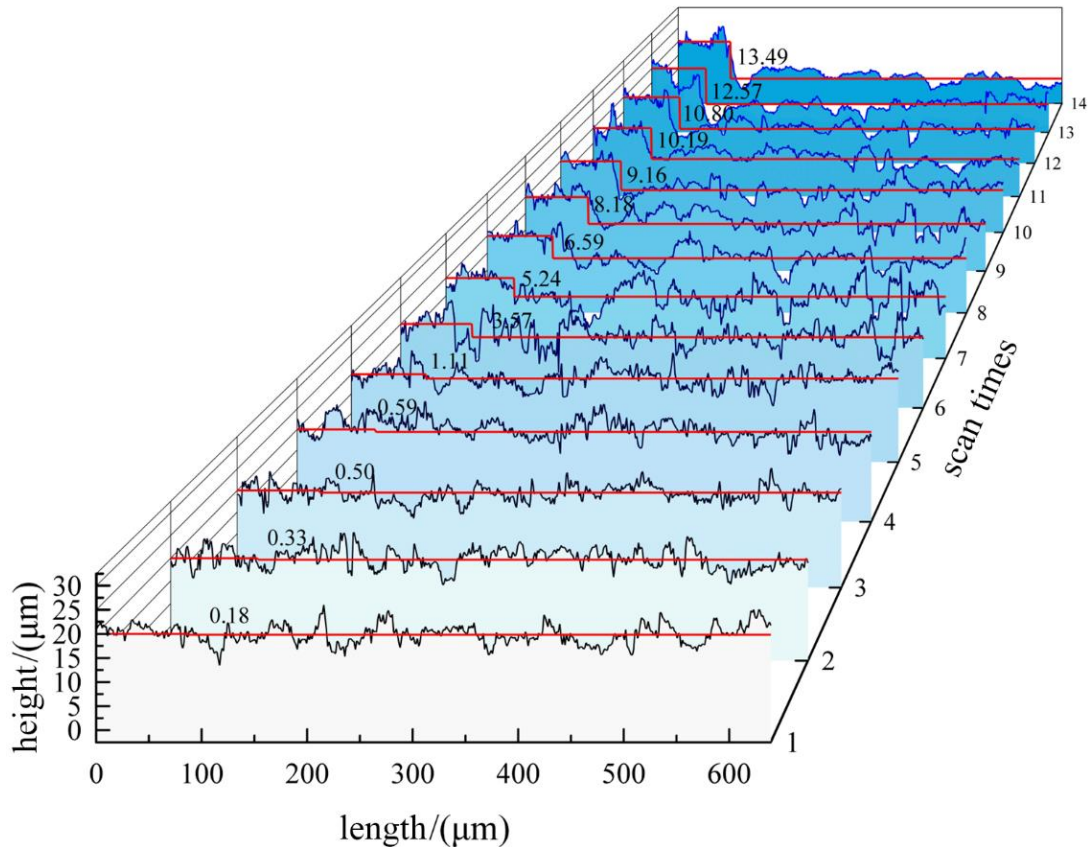


Fig.5. Cross-Sectional Height Difference of Sample after processing

3.3 PEEK removal mechanism

3.3.1 Morphological Evolution and Photo-Thermal Effects in PEEK Ablation

Due to the substantial roughness at the metal-substrate interface, there are significant thickness variations, resulting in both copper residues and exposed substrate following initial processing. Complete removal of the copper layer and clean ablation inevitably require some substrate material removal, leading to unavoidable substrate over-ablation. Studies indicate that PEEK begins to thermally decompose at 600 K, with thermal degradation reaching 40% above 900 K, resulting in carbonization. Given PEEK's low thermal conductivity, heat accumulation from the high repetition rate used to enhance processing efficiency in this experiment cannot be overlooked. Therefore, examining the physical and chemical characteristics of PEEK post-ablation is crucial to achieving an outcome that avoids damage or carbonization.

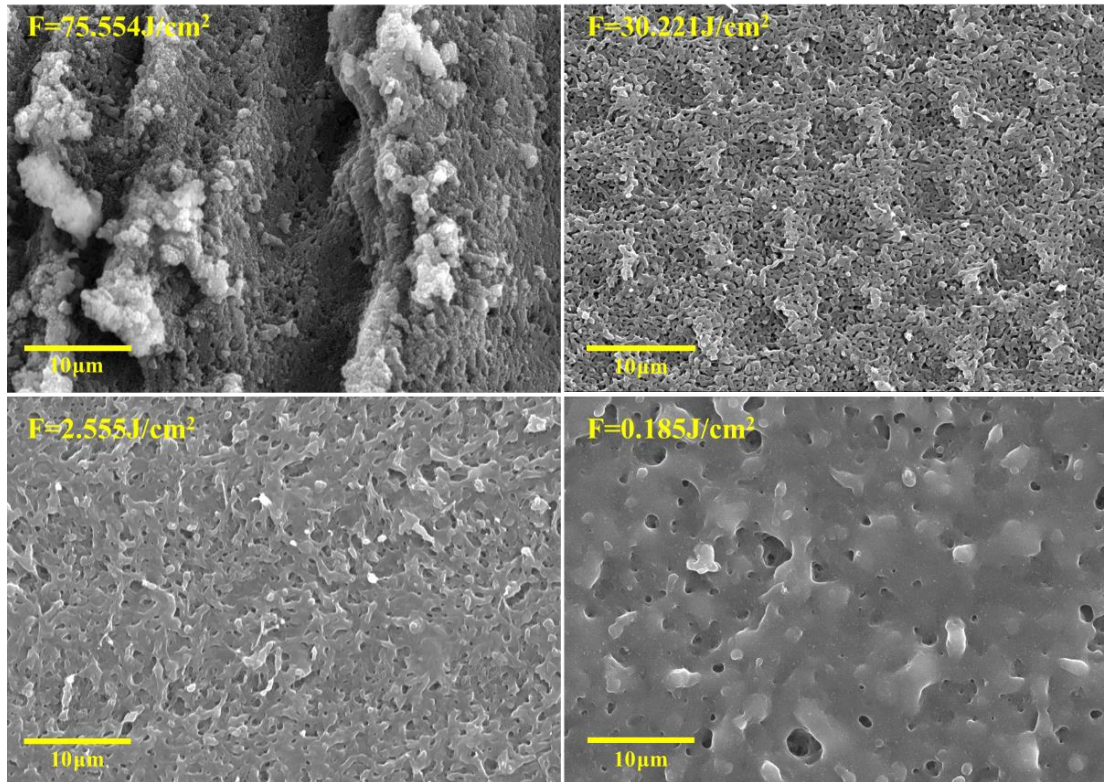


Fig.6. Morphological Changes of PEEK Substrate after Ablation at Different Fluences

The morphology, as shown in Fig.6, appears predominantly dark—black and gray—under optical microscopy. Electron microscopy reveals that, at high fluences, the substrate exhibits a compact, flake-like structure with clusters of high electron emissivity. This morphology likely results from excessive energy deposition, causing instantaneous melting of the PEEK surface and subsurface layers. Upon resolidification, the material crystallizes into fragmented microflakes with a sheet-like structure, accompanied by carbide formation. At fluences several times the ablation threshold or lower, the substrate presents a microscale, irregular, porous network. This network structure arises from the explosive boiling phenomenon during ablation, with microdroplets ejected and redeposited, while vapor recoil pressure overcomes surface tension and gravity to form a keyhole structure.

Conductivity tests indicate that the processed substrate is non-conductive, suggesting that the optical transformation of the material surface is not caused by carbonization from a thermophysical process. Instead, it is hypothesized that surface photo-oxidation occurs during ablation, altering the absorption spectrum of PEEK. Additionally, the heat-induced network structure enhances light-trapping effects, capturing and scattering incident light, which increases visible light absorption.

3.3.2 Raman Spectroscopy Analysis of PEEK Post-Ablation

To investigate the effects of femtosecond laser processing on PEEK substrates, XPS and Raman spectroscopy analyses were conducted on the processed samples. Polyether ether ketone (PEEK) is a high-molecular-weight polymer with long carbon chains, chemically represented as $(C_{19}H_{18}O_3)_n$. Under high-energy conditions, it undergoes dehydrogenation, deoxygenation, and bond reorganization, leading to the formation of carbides. Previous studies have shown that this process primarily generates carbonaceous structures such as amorphous carbon and graphene.

Using a 785 nm wavelength laser to excite PEEK suppresses fluorescence effects, allowing for the detection of Raman scattering associated with covalent bond vibrations in PEEK. Unprocessed PEEK exhibits four prominent characteristic peaks: 808 cm^{-1} , corresponding to the C - H bond in benzene rings; 1146 cm^{-1} , associated with the O - C - O ether bond vibration; and peaks at 1604 cm^{-1} and 1643 cm^{-1} , corresponding to the C=C double bond and carbonyl group vibrations.

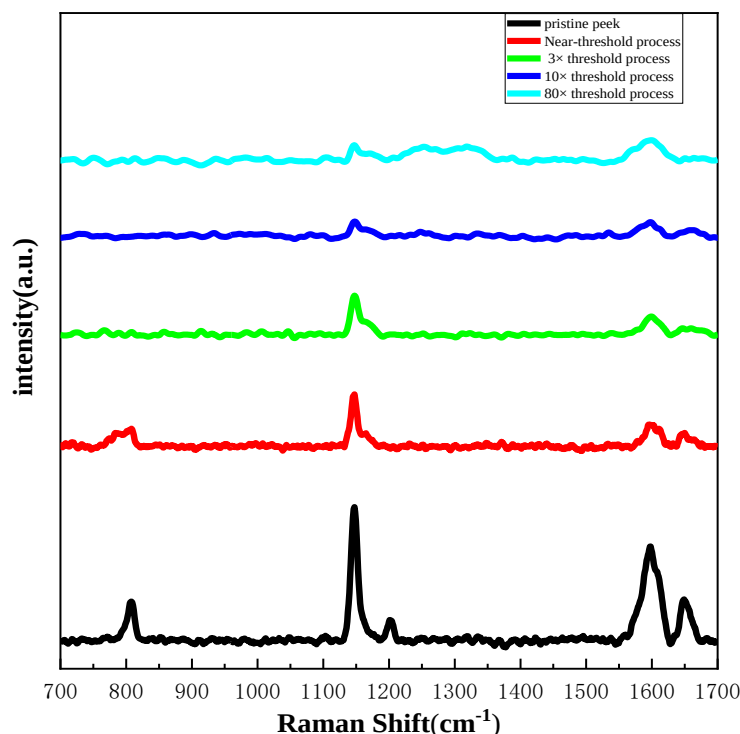


Fig.7. Raman Spectra of PEEK Processed at Different Energies under 785 nm Excitation

Fig. 7 shows the Raman spectra of PEEK processed at different energy levels under 785 nm excitation. A comparison of the irradiated and non-irradiated PEEK Raman spectra indicates that near-threshold laser processing at 1030 nm causes minimal changes in chemical composition. However, at three times the threshold energy, there is a significant reduction in C=O bonds, likely due to photochemical reactions during processing that reduce the highly oxidizing C=O bonds. This process also weakens the C=C and O - C - O bonds and broadens the spectral peaks, consistent with the morphology observed in electron microscopy. During ablation, PEEK undergoes explosive boiling, which is accompanied by random chain scission and high-temperature cross-linking, resulting in reduced crystallinity. At much higher fluences, the heat-affected zone extends deeper into the subsurface, where substantial energy accumulation creates a localized high-temperature, high-pressure environment. In this state, material removal and carbonization occur simultaneously, causing characteristic peaks to disappear, with only the benzene ring vibration and C=C double bond peaks remaining.

3.3.3 XPS Characterization of PEEK Surface Chemistry

As shown in Fig. 8 and Table 3, the XPS C1s spectra reveal that carbon in pristine PEEK primarily originates from

benzene rings and carbon-oxygen bonds, with approximately 64.92% in sp^2/sp^3 carbon, 20.98% in C – O bonds, and 8.77% in C=O bonds. Simultaneously, the presence of the $\pi-\pi^*$ satellite peak was observed, accounting for 5.34%. Studies on laser bonding of metals with PET have shown that carbonyl groups (C=O) in polymers, due to their high reactivity, readily convert to C – O – M or C – M at high temperatures at the metal interface [20, 21].

As indicated in Table 6, after near-threshold laser processing on PEEK, the total proportion of C – O bonds in the substrate remains unchanged, around 20%, while the C=O bonds decrease, suggesting a reduction of carbonyl groups during processing, likely converting C=O into C – O – C, as illustrated in Fig. 8b. With increased laser fluence, the proportion of C – C bonds rises, while C – O and C=O bonds correspondingly decrease. This indicates that energy deposition induces a temperature increase, promoting scission of long-chain oxygen bridges and high-temperature cross-linking in PEEK, leading to an increase in carbon content in the substrate but still insufficient to form a conductive carbon network. Even at fluences significantly exceeding the ablation threshold, the rapid temperature rise from instantaneous heating triggers explosive boiling, altering the subsurface morphology of PEEK and causing vaporization. Carbonization products are expelled with the shockwave generated by material removal, and the majority of energy dissipates through ablation cooling. As a result, the processed area largely retains the original PEEK composition.

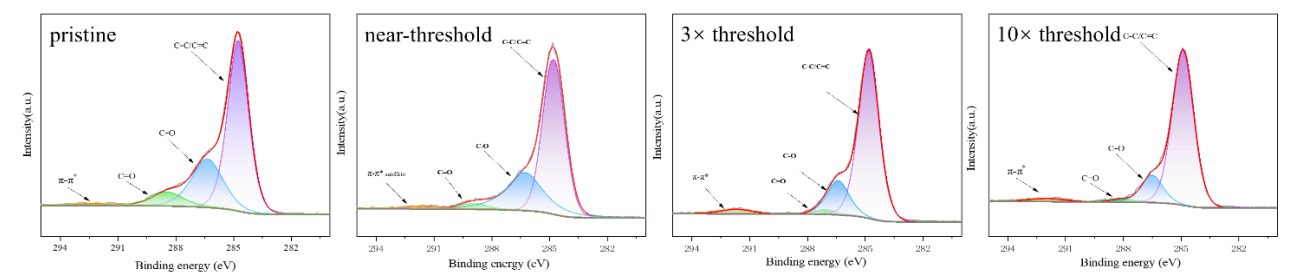


Fig. 8. XPS characterization of surfaces ablated at different fluences

Table 3. C1s regions of XPS

Type of C-element	Bind energy (EV)	Count (%)			
		pristine	near-threshold	3× threshold	10× threshold
C-C	284.8	64.92	68.17	71.47	74.54
C-O	286.5	20.98	19.80	17.39	15.73
C=O	288.6	8.77	6.52	5.81	5.03
$\pi-\pi^*$ satellite	~ 291	5.34	5.51	5.33	4.69

4.CONCLUSION

This study confirms the feasibility and effectiveness of femtosecond laser direct writing for precise patterning of Cu/Ni/Au-coated PEEK substrates, specifically targeting applications in high-performance, lightweight circuits and conformal antennas within the aerospace sector. Using predetermined laser parameters, selective ablation of the metal layer was achieved without inducing carbonization or structural damage to the underlying PEEK substrate.

However, the analysis of variable pulse widths revealed that at a wavelength of 1030 nm, selective ablation of copper without affecting the PEEK substrate is unachievable. Copper consistently exhibits a higher ablation threshold than PEEK, and the necessary surface roughness at the copper-PEEK interface to ensure adhesion further complicates this process, inevitably leading to partial ablation of the PEEK substrate during copper removal. Therefore, it is crucial to further investigate the post-ablation morphology and physicochemical properties of PEEK to identify optimal laser parameters that enable effective material removal while preserving the substrate's inherent properties.

In-situ LIBS monitoring provided accurate compositional and depth information, enabling precise detection of the copper-PEEK interface and supporting efficient material removal with minimal substrate damage. Subsequent SEM, Raman spectroscopy, and XPS analyses confirmed that optimized femtosecond laser parameters effectively minimize thermal damage, enhance ablation precision, and preserve PEEK's surface integrity.

These findings highlight femtosecond laser direct writing as a promising alternative to traditional multi-step fabrication methods, offering significant environmental advantages, enhanced resolution, and greater processing flexibility. This research provides a foundation for continued exploration and refinement of laser parameters for advanced polymer-metal composite processing in aerospace applications.

ACKNOWLEDGMENT

The authors would like to acknowledge the financial support from Sichuan Province Engineering Technology Research Center of Broadband Electronics Intelligent Manufacturing (BEIM2022B005) .

REFERENCES

- [1]Liu, L., et al., High-quality and efficient large-area copper removal utilizing laser-induced active mechanical peeling. *Optics Express*, 2023. **31**(25): p. 42191-42205.
- [2]Yang, G., et al., Insight into the surface behavior and dynamic absorptivity of laser removal of multilayer materials. *Optics Express*, 2023. **31**(23): p. 37483-37494.
- [3]Yang, G., et al., An adaptive laser processing method of multilayer heterogeneous materials by monitoring dynamic spectrum at kHz-MHz level. *Optics and Lasers in Engineering*, 2023. **170**: p. 107792.
- [4]Weber, R., et al., Processing constraints resulting from heat accumulation during pulsed and repetitive laser materials processing. *Optics express*, 2017. **25**(4): p. 3966-3979.
- [5]Gaitanelis, D., et al., A multi-technique and multi-scale analysis of the thermal degradation of PEEK in laser heating. *Polymer Degradation and Stability*, 2023. **211**: p. 110282.
- [6]Assaf, Y., M. Zhao, and A.-M. Kietzig, Femtosecond laser-induced porosity on poly (ethylene) surfaces—A crystallographic and rheological study. *Journal of Applied Physics*, 2018. **124**(2).
- [7]Baudach, S., J. Krüger, and W. Kautek, Femtosecond laser processing of soft materials. *The Review of Laser Engineering*, 2001. **29**(11): p. 705-709.
- [8]Guay, J., et al., Polarization-dependent femtosecond laser ablation of poly-methyl methacrylate. *New Journal of Physics*, 2012. **14**(8): p. 085010.
- [9]Lamberti, A., et al., All-SPEEK flexible supercapacitor exploiting laser-induced graphenization. *2D Materials*, 2017.

4(3): p. 035012.

- [10]Moon, H.-Y., et al., Dynamic changes in PDMS surface morphology in femtosecond laser treatment. *Optics Express*, 2015. **23**(15): p. 19854-19862.
- [11]Singh, S.S., et al., Incubation studies and the threshold for surface damage and cavity formation in the processing of polycarbonate by Nd: YAG laser. *Optics & Laser Technology*, 2018. **108**: p. 592-601.
- [12]Li, Q., et al., Femtosecond laser micro-structuring of amorphous polyether (ether) ketone at 775 nm and 387 nm. *Journal of Physics D: Applied Physics*, 2020. **53**(36): p. 365301.
- [13]Ouyang, Z., et al., Preparation of high-quality three-dimensional microstructures on polymethyl methacrylate surfaces by femtosecond laser micromachining and thermal-induced micro-leveling. *Optics & Laser Technology*, 2022. **145**: p. 107499.
- [14]Rahaman, A., A. Kar, and X. Yu, Time-resolved measurements of optical properties in ultrafast laser interaction with polypropylene. *Optics Express*, 2020. **28**(2): p. 2640-2648.
- [15]Rebollar, E., et al., Assessment of femtosecond laser induced periodic surface structures on polymer films. *Physical Chemistry Chemical Physics*, 2013. **15**(27): p. 11287-11298.
- [16]Winter, J., et al., Ultrashort single-pulse laser ablation of stainless steel, aluminium, copper and its dependence on the pulse duration. *Optics express*, 2021. **29**(10): p. 14561-14581.
- [17]Zayarny, D.A., et al., Pulse-width-dependent surface ablation of copper and silver by ultrashort laser pulses. *Laser Physics Letters*, 2016. **13**(7): p. 076101.
- [18]Ho, J., M. Olguin, and C. Diaz, Poly (aryl-ether-ether-ketone) as a possible metalized film capacitor dielectric: accurate description of the band gap through ab initio calculation. ARMY RESEARCH LAB ADELPHI MD SENSORS AND ELECTRON DEVICES DIRECTORATE, 2014.
- [19]Bolshakov, A., et al., Single crystal diamond growth by MPCVD at subatmospheric pressures. *Materials Today Communications*, 2020. **25**: p. 101635.
- [20]Liao, W., S. Zhao, and M. Gao, Oscillating laser conduction joining of dissimilar PET to stainless steel. *Polymers*, 2022. **14**(22): p. 4956.
- [21]Liu, F., et al., On formation of AlOC bonds at aluminum/polyamide joint interface. *Applied Surface Science*, 2019. **466**: p. 202-209.